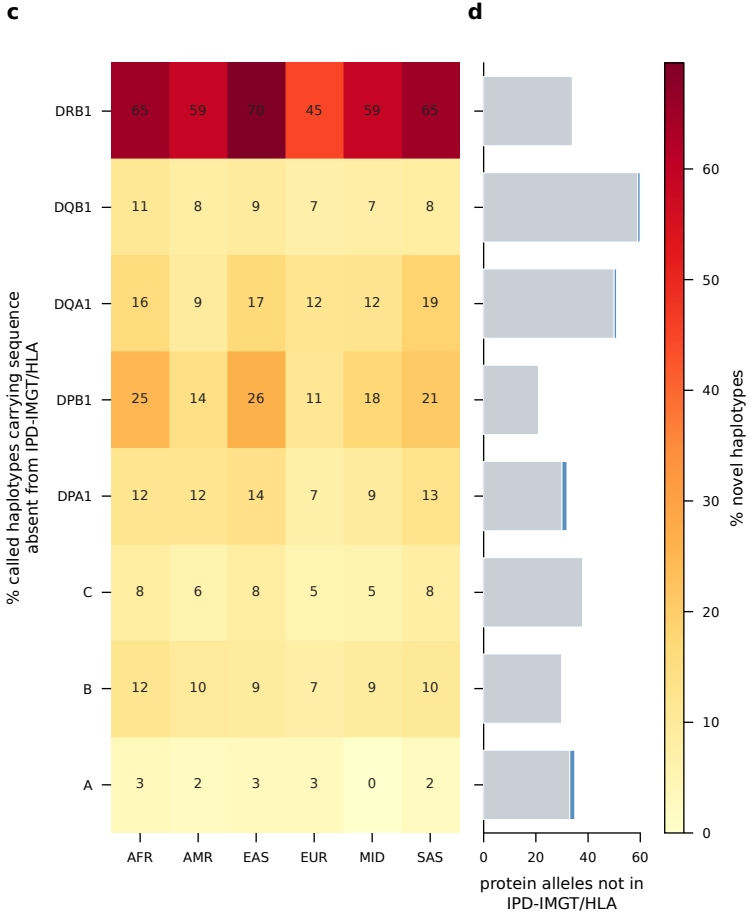
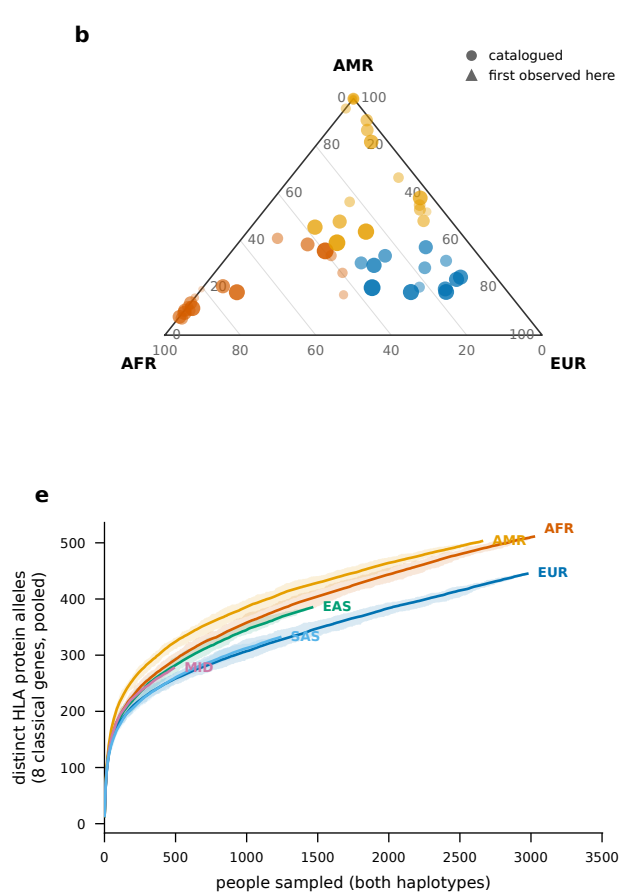
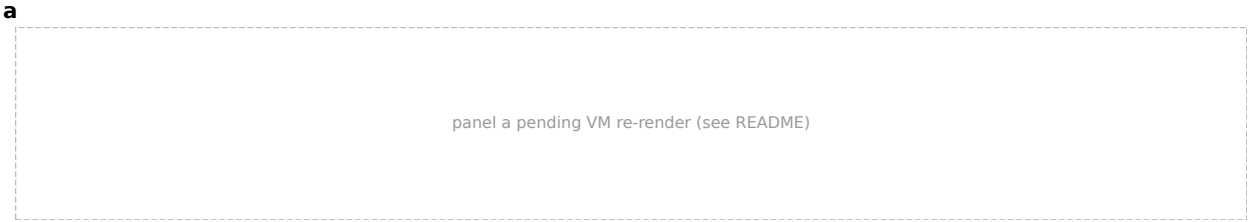




a all unrelated participants (n=11833), predicted ancestry, ordered least->most admixed within block; per-group % clearing the strict ancestry threshold (≥ 0.98) is in panel_a_group_sizes.tsv, not annotated on the panel.
b HLA-B alleles, strict ancestry (probability ≥ 0.98), ≥ 20 carriers; point area/opacity ~ carrier count (log). c exact lower-bound rate; hatched cell = denominator < 20 haplotypes. d pooled across ancestry (ancestry split: reports/hla_popgen/34_novel_recurrence/, all genes). e 25 permutations/point, mean $\pm$ 95% band, 'pred' ancestry scheme; see reports/hla_popgen/39_saturation_by_ancestry/ for the strict95 sensitivity curve.

seen once
2-19 unrelated people
 ≥ 20 unrelated people